🔧 Engine Teardown & Rebuild

Small Engines Overhaul · Unit 8 · Level 2



Name: _____

Modified from the class “Engine project + Engine Assembly 1 + Ring End Gap Handout”.

🧺 Tools

- ☐ 🧰 **Socket set + wrenches**
- ☐ 🔘 **Flywheel puller** with adult
- ☐ 🔧 **Torque wrench** with adult
- ☐ 🌀 **Valve spring compressor** with adult

🧺 Supplies

- ☐ 🥣 **Parts tray** labeled cups
- ☐ 📄 **Feeler gauges**
- ☐ 🔄 **Lapping compound**
- ☐ 🗒️ **New gasket set**

🔧 Steps: check each box when you finish

- [ ] 1 🔌 Disconnect plug wire; with adult supervision drain fuel and oil.
- [ ] 2 📸 Photo; remove shroud, air filter, carb, muffler (tray cups 1-4).
- [ ] 3 🛞 Remove flywheel with the puller; remove armature.
- [ ] 4 🔩 Remove head; remove valves with the spring compressor.
- [ ] 5 ⭕ Remove piston/rod; check ring end gap and side clearance; record.
- [ ] 6 🔄 Lap valves; set valve clearance to spec.
- [ ] 7 ⏱️ Reassemble: align timing marks; new gaskets; oil parts.
- [ ] 8 🔧 Torque head bolts in pattern to spec.
- [ ] 9 🔍 Test: fuel, spark, compression; adult starts the engine outdoors.

🚧 **Under construction:** The class engine project steps and rubric are not written in Canvas; this checklist is team-written. Specs vary by engine model; staff fill them in from the class manual. *Check with the class teacher.*